where $\boldsymbol{J}$ is the skew symmetric symplectic structure matrix and

$$\tilde{H}_h = H + h^r H_r + h^{r+1} H_{r+1} + \dots \tag{3.10}$$

We have worked out formulas above for the first terms in the expansion in the case of several splitting methods. We can thus view (at least locally) our numerical trajectory as the stroboscopic map (snapshots) of the evolution of a continuous Hamiltonian system.

Ideally, we would like to be able write that the solution to the modified equations is equivalent to mapping via the specified one-step method:

$$\mathscr{G}_h = e^{h\mathcal{L}_{\tilde{f}_h}}.$$

Unfortunately, this statement is elusive. The series expansion in question rarely converges. It is easy to show that for the harmonic oscillator, the series does indeed converge, but it turns out that this is an anomaly. The reason for the lack of convergence of the series is, precisely, its global nature (which is also its chief benefit). In order to proceed, one normally has to truncate the series at some finite number of terms. A natural question is where should we terminate the series? If we include only the $\mathcal{O}(h^r)$ term, then we have not really gained much. For example we would not be able to say that $\tilde{H}_h$ is essentially a conserved quantity along the numerical solutions.

The analysis of the expansion was first treated systematically in [31, 359], where it was shown that a *finite* truncation to terms of order h^k can be achieved in such a way that the error is "exponentially small" in the stepsize. (See also [163, 310].) Define

$$\bar{H}_k \stackrel{\text{def}}{=} H + h^r H_r + h^{r+1} H_{r+1} + \dots + h^k H_k. \tag{3.11}$$

Let us state and prove the following basic result (see [162, 164, 227]):

Theorem 3.1 *Suppose the Hamiltonian H and modified Hamiltonian $\bar{H}_k$ are smooth functions globally defined on a convex, compact subset $\mathcal{B}$ of $\mathbb{R}^{2N_c}$ and suppose that the exact solution and numerical approximations (for h sufficiently small) are confined to $\mathcal{B}$. Then, we have asymptotically for $h \to 0$ that*

$$H(\boldsymbol{z}_n) = H(\boldsymbol{z}_0) + \mathcal{O}(h^r),$$

for $n = 0, 1, \dots, \nu$ where $\tau = \nu h = \mathcal{O}(h^{-k+r})$.

Proof First observe that due to smoothness and the assumptions on $\mathcal{B}$, we have a global Lipschitz constant

$$|H(\boldsymbol{u}) - H(\boldsymbol{v})| \leq L\|\boldsymbol{u} - \boldsymbol{v}\|$$

for all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{B}$.

Denote by $\mathscr{F}_h^{(k)}$ the flow map of the truncated Hamiltonian expansion $\bar{H}_k$,

$$\mathscr{F}_h^{(k)} \stackrel{\text{def}}{=} \exp(h\mathcal{L}_{\bar{H}_k}).$$

By construction, we have

$$z_{n+1} = \mathscr{F}_h^{(k)}(z_n) + \eta_n$$

where $\|\eta_n\| \le Ch^{k+1}$.

Since $\mathscr{F}_h^{(k)}$ preserves its Hamiltonian (3.11), we have

$$\bar{H}_k(\mathscr{F}_h^{(k)}(z_n)) = \tilde{H}_k(z_n).$$

Now

$$\begin{aligned}
\bar{H}_k(z_\nu) - \bar{H}_k(z_0) &= \sum_{n=0}^{\nu-1} \bar{H}_k(z_{n+1}) - \bar{H}_k(z_n) \\
&= \sum_{n=0}^{\nu-1} \bar{H}_k(z_{n+1}) - \tilde{H}_k(\mathscr{F}_h^{(k)}(z_n)) \\
&= \sum_{n=0}^{\nu-1} \bar{H}_k(\mathscr{F}_h^{(k)}(z_n) + \eta_n) - \tilde{H}_k(\mathscr{F}_h^{(k)}(z_n)).
\end{aligned}$$

Hence

$$|\bar{H}_k(z_\nu) - \bar{H}_k(z_0)| \le L \sum_{n=0}^{\nu-1} \|\eta_n\| \le L\nu h^{k+1}.$$

We have $\nu = \tau/h$, thus

$$|\bar{H}_k(z_\nu) - \bar{H}_k(z_0)| \le L\tau h^k.$$

Next observe that

$$H = \bar{H}_k - h^r H_{(r)} - h^{r+1} H_{(r+1)} - \ldots - h^k H_{(k)} = \bar{H}_k + \mathcal{O}(h^r).$$

Therefore

$$|H(z_\nu) - H(z_0)| \le L\nu h^{k+1} + \mathcal{O}(h^r),$$

so that, as long as $\nu \leq C_2 h^{-k+r-1}$, we have

$$|H(z_\nu) - H(z_0)| \leq \mathcal{O}(h^r).$$

□

If the differential equations are infinitely differentiable, we may take the truncation index k as large as we like, but the constants appearing in the above theorem will depend on the truncation index in a complicated way. It is possible to prove (see discussions in [164, 227] for more detail), that for many standard classes of numerical methods, there are real, positive constants C, D such that

$$\|\mathscr{G}_h(\boldsymbol{\zeta}) - \mathscr{F}_h^{(k)}(\boldsymbol{\zeta})\| \leq Ch\,[D(k+1)h]^{k+1}\,,$$

giving a precise bound on the magnitude of the difference between the time h evolution under the truncated perturbed Hamiltonian and the numerical method.

If h is small, then for k sufficiently small the quantity in brackets is less than one and the difference from the truncated approximation decreases in magnitude with increasing k. As soon as k satisfies

$$k + 1 > \frac{1}{Dh}$$

the power grows monotonically without bound. We can minimize the difference between $\mathscr{G}_h$ and $\mathscr{F}_h^{(k)}$ by choosing

$$k = \frac{1}{hDe} - 1,$$

in which case,

$$\|\mathscr{G}_h(\boldsymbol{\zeta}) - \mathscr{F}_h^{(k)}(\boldsymbol{\zeta})\| < \tilde{C}he^{-\frac{1}{\gamma h}}, \qquad \gamma = \frac{1}{De}.$$

This bound tends to zero extremely rapidly (more rapidly than any power of h) as $h \to 0$. The assumption is that if this error in the approximation of the numerical map by the flow of a truncated perturbed Hamiltonian flow map is much smaller than the other errors present in our model, any drift due to lack of convergence would be minor, perhaps even invisible, on the timescale of simulation.

It is not recommended to take these estimates too literally, i.e. to try to compute the constants C and D (which might be quite difficult). Instead, we simply use the estimates to guide our intuition that the energy will be expected to fluctuate on very long time intervals with a magnitude bounded by $\mathcal{O}(h^r)$. To gain a full understanding of the issue of energy drift in the setting of molecular dynamics, the article [121] is highly relevant.